

***Dialäkt Äpp*: communicating dialectology to the public – crowdsourcing dialects from the public**

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Abstract

The smartphone application *Dialäkt Äpp* capitalizes on the Swiss' public interest in dialectology and features the following core functionalities: (1) users indicate their pronunciation variants of 16 words and the application predicts their local dialect, (2) users record their pronunciation of the same 16 words, which are then uploaded on a server and displayed on an interactive map. In this sense the app offers an auditory journey across the Swiss dialectal landscape. The primary goal of the free of charge iOS application is science communication to a broad public: *Dialäkt Äpp* is appealing to the user and has been downloaded quite frequently (>58'000, January 2014). This novel approach also benefits the researcher: New media allow for new data collection methods in a number of fields. In linguistics, for example, crowdsourced data has mainly been obtained through online surveys. With *Dialäkt Äpp*, we present a novel method of crowdsourcing linguistic variation by means of a smartphone application: with the users' consent, we obtain pronunciation data of 16 words for thousands of dialect speakers stemming from all over German-speaking Switzerland. This corpus bears potential for studies on regional variation, linguistic change, and population statistics, the latter of which has implications for forensic phonetics.

Keywords

automatic dialect recognition, crowdsourcing, iOS smartphone application

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1.0 Introduction

We developed an innovative method that allows us to communicate dialectology to a broader public on the one hand and to crowdsource dialect data on the other: meet the iOS application *Dialäkt Äpp* (Leemann & Kolly, 2013; Wemakeit, 2012). Crowdsourcing refers to “the practice of obtaining needed [...] content by soliciting contributions from a large group of people and especially from the online community [...]” (Merriam-Webster, 2014).

How did the app come about? At a cocktail party in Switzerland it is common to hear somebody ask “You just said [ˈxæuə] for *Kelle* – do you come from Bern?” Observations like this reveal that Swiss German (SwG) speakers are well aware of geographical variation of their dialects; for example, they usually know that /l/-vocalization is typical for the Bernese region. Such anecdotal evidence spurred us to develop *Dialäkt Äpp*, a smartphone application that capitalizes on the Swiss’ keen awareness of and interest in linguistic variability. The app provides high user appeal and has two main functionalities: (1) to predict the users’ dialect based on their selection of dialectal variants of 16 words presented in Standard German, (2) to allow users to record their pronunciation of the same words, upload the resulting audio files on a server and display them on an interactive map. In this sense the app offers an auditory journey through the SwG dialectal landscape. The corpus generated over the past 10 months by this free of charge iOS application includes variant selection data (data retrieved through functionality (1)) from almost 42’000 speakers and audio recording data (data retrieved through functionality (2)) from almost 2600 speakers from all over German-speaking Switzerland. The app has been downloaded over 58’000 times (as of January 13, 2014; app release: March 22, 2013), was the number one downloaded free app for iOS devices in Switzerland (App Annie, 2013), and gained major media attention in the German- as well as French-speaking parts of Switzerland (cf. <http://www.pholab.uzh.ch/leute/leemann/app.html>).

Why is *Dialäkt Äpp* an interesting tool for collecting and analyzing linguistic data? Firstly, the iPhone microphones cover a wide frequency range of 50–20’000 Hz that enable high-quality audio recordings (Faber Acoustical, 2009). Previous research showed that audio data collected with a first generation iPhone from 2007 allows for reliable acoustic measurements of vowel formants that can be used for speech analyses (cf. de Decker & Nycz, 2011). Secondly, the positive resonance from the public shows that users enjoy this app, which propelled the number of speakers that donated their voices for the app. This confirms findings by Parent & Eskenazi (2011): a higher number of participants will crowdsource data if they are provided with playful, appealing content. Thirdly, it has been shown in research on crowdsourcing data that the input of most users is reliable, and the amount of data compensates for individual flaws: data redundancy accounts for its quality (Munro, Bethard, Kuperman, Lai, Melnick et al., 2010; Juska-Bacher, Biemann, Quasthoff, 2012). Data crowdsourced by *Dialäkt Äpp* thus bears significant scientific potential. For example, the *Dialäkt Äpp* corpus allows for an area-covering study of linguistic change in the SwG dialect landscape: our dialect prediction algorithm is based on historical maps of the *Sprachatlas der Deutschen Schweiz* (Linguistic Atlas of German-speaking Switzerland, SDS, 1962–2003). Through its main functionalities, *Dialäkt Äpp* collects pronunciation data of the same localities used in the SDS-maps, therefore enabling a direct comparison with historical data.

This paper is structured as follows: Section 2 focuses on the user by introducing the functionalities of *Dialäkt Äpp* as well as the methods used to implement them. Section 3 focuses on the researcher in discussing the scientific potential of the corpus crowdsourced through *Dialäkt Äpp*. We round off this section by presenting preliminary research conducted with this corpus.

2.0 *Dialäkt Äpp* for the user: functionalities and methods

Dialäkt Äpp features the following four functionalities:

- Dialect prediction (original functionality name in the app: *Verortung*)

- Localities and their pronunciations – pronunciations and their localities (*Wie? Wo?*)
- Audio recordings (*Aufnahme*)
- Word of the week (*Wort der Woche*)

2.1 Dialect prediction

“What dialect do you speak?” – a frequent question at social events in German-speaking Switzerland, and *Dialäkt Äpp* provides an answer to this question, the validity of which can be assessed by the user.

Functionality: For 16 words, *Dialäkt Äpp* prompts its users to select their pronunciation variant from a list by tapping on the smartphone screen. Since there is no standardized SwG orthography, variants are spelled close to pronunciation. We further provided phonetic transcriptions in cases where variants cannot be differentiated by spelling (e.g. *Donschtig* [ɔ], *Donschtig* [o]; cf. Figure 1, left). Since users may not be familiar with phonetic transcription, the app further offers pre-canned recordings of all variants. Phonetic transcriptions and pre-canned recordings allowed us to include a large number of variants for each variable, which should refine the accuracy of dialect prediction. Each variable with its variants is presented on a single screen with the Standard German word in its title and the relevant variable (e.g. *Erster Vokal* ‘first vowel’) in its subtitle. Variables are mostly phonetic, but we also included one lexical and one morphological variable. Below the subtitle, pronunciation variants are listed, where the relevant variable is underlined (cf. Figure 1, left). Table 1 presents the 16 variables.

Table 1: Variables chosen for dialect prediction (MHG = Middle High German), examples, numbers and types of SwG variants (phon. = phonetic, lex. = lexical, morph. = morphological)

Standard German word	Variable	Examples of variants	N	Type
<i>Abend</i> ‘evening’	MHG <i>-ent</i> , <i>-ant</i> , <i>-int</i> , <i>-unt</i>	<i>Aabe</i> , <i>Aabig</i>	13	phon.
<i>Abend</i> ‘evening’	MHG <i>â</i>	<i>Aabe</i> [a], <i>Aabe</i> [a]	8	phon.
<i>Apfelüberrest</i> ‘apple core’	lexeme	<i>Bütschgi</i> , <i>Gröibsch</i>	39	lex.
<i>Augen</i> ‘eyes’	MHG <i>ou</i>	<i>Aige</i> , <i>Ouge</i>	11	phon.
<i>Bett</i> ‘bed’	MHG <i>e</i>	<i>Bett</i> [e], <i>Bett</i> [ɛ]	2	phon.
<i>Donnerstag</i> ‘Thursday’	MHG <i>o</i>	<i>Donschtig</i> [o], <i>Dunschtig</i> [u]	8	phon.
<i>Kelle</i> ‘ladle’	MHG <i>ll</i>	<i>Chäle</i> , <i>Chäue</i>	5	phon.
<i>Kind</i> ‘child’	MHG <i>k</i> , <i>ch</i>	<i>Chind</i> [x], <i>Schind</i> [ʃ]	4	phon.
<i>Tanne</i> ‘fir’	MHG <i>n</i> , <i>nn</i>	<i>Tane</i> , <i>Tanne</i>	2	phon.
<i>fragen</i> ‘to ask’	MHG <i>â</i> , <i>æ</i>	<i>fraage</i> , <i>frääge</i>	10	phon.
<i>heben</i> ‘to lift’	MHG <i>u</i>	<i>lupfe</i> [u], [ʊ], <i>löpfe</i> [y], [ʏ]	3	phon.
<i>hinauf</i> ‘upwards’	MHG <i>hin-ûf</i>	<i>embruf</i> / <i>embrüf</i> , <i>uehi</i>	31	morph.
<i>schneien</i> ‘to snow’	MHG <i>î</i>	<i>schneie</i> [ɛ], <i>schniie</i>	7	phon.
<i>spät</i> ‘late’	MHG <i>â</i> ,	<i>schpaat</i> [ɔ], <i>schpeät</i> [eə]	12	phon.
<i>tief</i> ‘deep’	MHG <i>iu</i>	<i>töif</i> , <i>tüüf</i>	7	phon.
<i>trinken</i> ‘to drink’	MHG <i>î</i> , <i>-nk-</i>	<i>treeche</i> , <i>trinkche</i>	10	phon.

After going through the 16 variables, the algorithm presents users with a list of five localities that best correspond to their set of pronunciation variants (Figure 1, center left), and displays them on a map (Figure 1, center right). Users can assess the validity of the first hit (Figure 1, center left): if they accept the result, users are asked to indicate their age and gender for scientific purposes (Figure 1, right), and thereby support our research. If they feel the result is not accurate, users can specify their dialect by choosing from a list of localities. Users’ pronunciation variants together with this metadata are anonymously stored on a server.



Figure 1: Pronunciation variants for *Donnerstag* ‘Thursday’ (left), dialect prediction displayed as a list of five best hits (center left) and as pins on a map (center right), prompt that asks the users to release their data for scientific purposes (right)

Method: The dialect prediction algorithm is based on 16 maps of the *SDS* (1962–2003), one of the most significant works on SwG dialectology. The *SDS* documents regional variation of SwG dialects in the middle of the 20th century – data were mainly collected in the 1940s (as was common at the time, elderly speakers were chosen as informants to obtain ‘older’ dialectal variants) and cover phonetic, phonological, morphological, lexical, and syntactic variation in 566 localities of German-speaking Switzerland. The number of maps selected, 16, underlies a trade-off between usability and prediction accuracy. As for usability, we assumed that more than 16 variables would result in the opting out of many users. On the other hand, we assumed that the accuracy of dialect prediction increased the more variables are included until prediction accuracy reaches saturation, i.e. that, after a certain number of variables, additional maps would not improve prediction. We intuitively felt that 15–20 maps fulfilled these criteria. Since the *SDS* documented dialects spoken around 1900–1950, we chose maps with isoglosses that were suspected to have remained stable. Previous research by Juska-Bacher (2010) showed that isoglosses of lexical variables underwent major changes. For this reason we mainly selected phonetic variables (cf. Table 1). We further chose maps with maximally small intersections. The map for *Bett* ‘bed’, for example, splits SwG in western and eastern dialects (Figure 2, left), whereas the map for *schneien* ‘to snow’ creates a northern and a southern area (Figure 2, center). The intersection of these two maps is a refined partition, as compared to each of the maps. A further intersection with the map for *tief* ‘deep’ creates an even higher number of subsets (Figure 2, right).



Figure 2: Maps for *Bett* ‘bed’ (left), *schneien* ‘to snow’ (center), *tief* ‘deep’ (right), cf. Glaser, Christen & Friedli (2013)

The 16 maps were converted into a table that forms the basis of our dialect prediction algorithm. The table contains a row for each locality, and a column for each pronunciation variant. Each cell features either a 1 (variant is present in the locality) or a 0 (variant is absent in the locality) (Table 2). The dialect prediction algorithm aggregates all the 1s for each row and coughs up the

locality with the highest aggregation as best hit. We adopted the localities used in the *SDS* and adjusted localities that were merged or have changed their names in the meantime, which currently results in 550 localities (BFS, 2012).

Table 2: Exemplary tabular basis for the dialect prediction algorithm. Rows document the pronunciation variants present in each of the 550 localities (1 = variant present, 0 = variant absent).

		German word	Abend (Endung)	Abend (Endung)	Abend (Endung)	Abend (Endung)
City	Canton	ID / Label	Aabig	Aabid	Aabe	Aaben
Aarau	Aargau	34	0	0	1	0
Aarberg	Bern	31	0	0	1	0
Aarburg	Aargau	45	0	0	1	0
Aarwangen	Bern	10	0	0	1	0
Adelboden	Bern	104	0	0	1	0
Aesch BL	Basel-Landsch	7	0	0	1	0
Aesch LU	Luzern	9	1	0	0	0

2.2 Localities and their pronunciations – pronunciations and their localities

Following the dialect prediction, users may be interested in the set of pronunciation variants that belongs to a certain locality.

Functionality: *Dialäkt Äpp* allows users to select a locality and see and hear the corresponding pronunciation variants for the 16 variables presented in Table 1. Furthermore, users may wish to explore where a certain pronunciation variant, such as vocalized /l/ in ['xæuə] *Kelle* 'ladle', is used. After the selection of a variable and its variant, an interactive map shows purple pins for all localities where the variant is present (cf. *SDS*, 1962–2003).

Method: This functionality is also based on the table used for dialect prediction (Table 2). The algorithm either extracts, for a chosen locality (row), all variants coded by 1, or it extracts, for a chosen variant (column), all localities coded by 1 and displays them on a map.

2.3 Audio recordings

Dialäkt Äpp invites its users to record their own pronunciation variants of the 16 variables and displays them on an interactive map. Users can enjoy an auditory journey through the SwG dialectal landscape.

Functionality: Users are instructed to record their voice in a maximally quiet environment (Figure 3, left). Prior to the recordings, users choose their dialect from a list of localities and indicate age and gender (Figure 3, center left). Each word then appears on a single screen, where an audio recording can be made, listened to, and re-recorded if needed. Users can further record their own dialectal rendition of a short dialogue (the first two turns of the *Gespräch am Neujahrstag*). An interactive map displays all the available recordings (Figure 3, center right). Purple pins indicate localities with contemporary user recordings, green pins indicate localities that feature historical (from 1943 and 1969) recordings of the *Gespräch am Neujahrstag* (cf. Phonogrammarchiv der Universität Zürich, 2000). Users can zoom in, tap on the pin that belongs to a locality, choose a variable and listen to all the recorded variants for this specific locality (Figure 3, right).

Method: Users' audio recordings and metadata are anonymously uploaded on a server and displayed on an interactive map. 24 historical recordings of the *Gespräch am Neujahrstag* appear on the same map (Figure 4, center right).



Figure 3: Instructions for recording (left), screen that prompts users to specify metadata (center left), localities for which contemporary (purple pins) and historical (green pins) recordings are available (center right), exemplary audio recordings of *Donnerstag* ‘Thursday’ for Zurich (right)

2.4 Word of the week

Many smartphone applications are used only once and subsequently deleted off the device (25%; cf. MobiThinking, 2013). To hopefully counteract this we implemented a functionality that incentivizes users with app content that changes over time. This is attractive because it makes users re-use the app and not delete it right away. In this way *Dialäkt Äpp* is interesting to use in the long run.

Functionality: Every Monday, a short text specifically crafted for non-linguists appears in the app. This short text contains etymological, diatopic and further information about one specific SwG word (Landolt, 2012–2013).

Method: The texts have kindly been provided by the Swiss German Dictionary (*Schweizerisches Idiotikon*).

3.0 *Dialäkt Äpp* for the researcher: Crowdsourced variant selection and audio data

Traditional methods for empirical linguistic research base their analyses on mostly small sets of speakers. Even recent rapid and anonymous studies typically reach a maximum of a few hundred participants (Mather, 2012; Leemann, Kolly, Werlen, Britain, & Studer-Joho, submitted). To obtain larger numbers of participants, linguists began using web-based technologies roughly about 10 years ago (Siebenhaar, 2003; Vaux & Golder, 2003; Schlippe, Ochs, & Schultz, 2013) – the largest sample obtained for SwG dialects includes over 14’000 speakers (Glaser, 2008). Recently, researchers in fields such as the cognitive sciences (Dufau, Duñabeita, Moret-Tatay, McGonigal, Peeters et al., 2011), or geophysics (Overeem, Robinson, Leijnse, Steeneveld, Horn et al., 2013) have efficiently used smartphone applications as tools to crowdsource even larger data sets. In the field of linguistics, the use of this technology is new: smartphone applications have been used to collect speech as a means to train acoustic models for automatic speech recognition (de Vries, Davel, Badenhorst, Basson, de Wet et al., 2014) or to document endangered languages (Iwaidja Inyman Team, 2012).

Two kinds of data are crowdsourced through *Dialäkt Äpp*: The dialect prediction functionality collects a set of variant selection pronunciation data of 16 words per user, as well as users’ age, gender and self-assessed dialect. The recording functionality collects a set of spoken data of maximum 16 words and two dialogue turns per user, as well as the same metadata. The corpus (as of January 13, 2014) comprises the variant selection data of 41’881 users (58% male, 42% female) and audio recordings of 2598 users (54% male, 46% female). For both kinds of data, most users are aged between 17 and 41 (interquartile range), and most users stem from the cantons of Zurich, Bern, Aargau, St. Gallen and Luzern (over 65%). We observe area-wide coverage: variant selection data were obtained for 547, audio data for 428 out of 550 localities, with often over 100 users per locality for the variant selection data. A preliminary evaluation of our dialect prediction algorithm reveals that the first locality predicted by the algorithm lies in the canton (administrative entity, but also dialect region) of the locality indicated by the user in 64% of the cases.

This data can be used by dialectologists to investigate between-locality – i.e. areal – variation, and by forensic phoneticians to create models of within-locality – e.g. between-speaker – variation.

With the *Dialäkt Äpp* corpus, dialectologists are now able to study the areal distribution of variants and create maps of areal variation, or quantify distances between different dialect regions, such as it has been done for American English (Katz, 2013) and German (Lameli, 2004; Elspass & Möller, 2003–2014) dialects. The *Dialäkt Äpp* data can additionally be compared to areal distributions documented in the *SDS*. This enables a comparison of the dialectal landscape documented in the 1940s to that of 2013 and thus to study sound change phenomena. In particular, the *Dialäkt Äpp* corpus allows a systematic study of the stability of phonetic, lexical and morphological isoglosses based on a small number of variables. A preliminary analysis on the variant selection data of the 41'881 users reveals that the dialectal variants for the lexeme *Apfelüberrest* 'apple core' and for *hinauf* 'upwards' have undergone more geographical change – e.g. the Zurich variant Bütschgi for *Apfelüberrest* has gained terrain – than the dialectal variants for the vowel in *tief* 'deep' and the gemination of /n/ in *Tanne* 'fir tree': based on a comparison of the *Dialäkt Äpp* corpus and the data documented in the *SDS*, we found that for *Apfelüberrest* and *hinauf*, only 53% and 58% of users respectively still use the variants documented in the *SDS*. For the phonetic variables in *tief* and *Tanne*, 69% and 83% of users respectively still use the pronunciation given in the *SDS*. Preliminary results are thus in line with Christen (1998) and Juska-Bacher (2010): phonetic variables seem to be more stable over time than lexical and morphological ones.

For forensic purposes, acoustic analyses of *Dialäkt Äpp* recordings allow unprecedented insights on the areal distribution of speech signal parameters such as fundamental and formant frequencies, temporal characteristics of segments, and speaking rate. Research in the field of forensic phonetics requires population data from a large set of speakers that feature little to no regional or social variation. For now, such population statistics only exist for certain languages (English, Standard German), and typically include around 50-100 speakers (Künzel, Masthoff & Köster, 1995; Jessen, 2007). A preliminary analysis on *Dialäkt Äpp* recordings of 115 users from the city of Bern and 205 users from the city of Zurich revealed that Bern SwG speakers talk significantly slower than Zurich SwG speakers – durational information contained in a few words alone allows to tell apart the two dialects. This common perceptual stereotype is thus mirrored in our production data (cf. Leemann, Kolly, Dellwo, submitted).

Hypothetically, a smartphone application like *Dialäkt Äpp* has the potential to crowdsource data for research on morphology, syntax, semantics, pragmatics, perceptual dialectology, language attitudes and many other fields. One could, for example, elicit data for research on language attitudes as well as dialect recognition by presenting users with a dialect quiz in which they have to attribute audio samples of different dialects to the corresponding regions and rate the samples on an affective scale (cf. Werlen, 1985).

4.0 Conclusions and outlook

Dialäkt Äpp is an interesting tool as a means to communicate dialectology to a large number of users from all over German-speaking Switzerland. At the same time it offers a vast amount of data for research in a number of fields. The data collected by *Dialäkt Äpp* also constitutes a unique scientific-historical collection of contemporary SwG dialects.

We have been approached by linguists who are interested in porting the *Dialäkt Äpp* framework to other platforms and create dialect prediction applications for dialects of other languages (American English, Italian, and German). In a follow-up project at our lab we are currently developing an app that predicts the user's dialect based on automatic speech recognition (*Swiss VoiceApp*, cf. Hove, Leemann, Kolly, Dellwo, Goldman, in this volume). Furthermore, we are currently using the area-covering variant selection data to create 16 contemporary maps of areal variation – one for each variable –, which can subsequently be compared to their counterparts from the *SDS* in order to study linguistic change in SwG. For example, one could examine the diffusion of /l/-vocalization in a dense net of localities, using the data on the variable *Kelle* – an analysis that has already been conducted for the Bernese region (cf. Hasler, 2013).

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